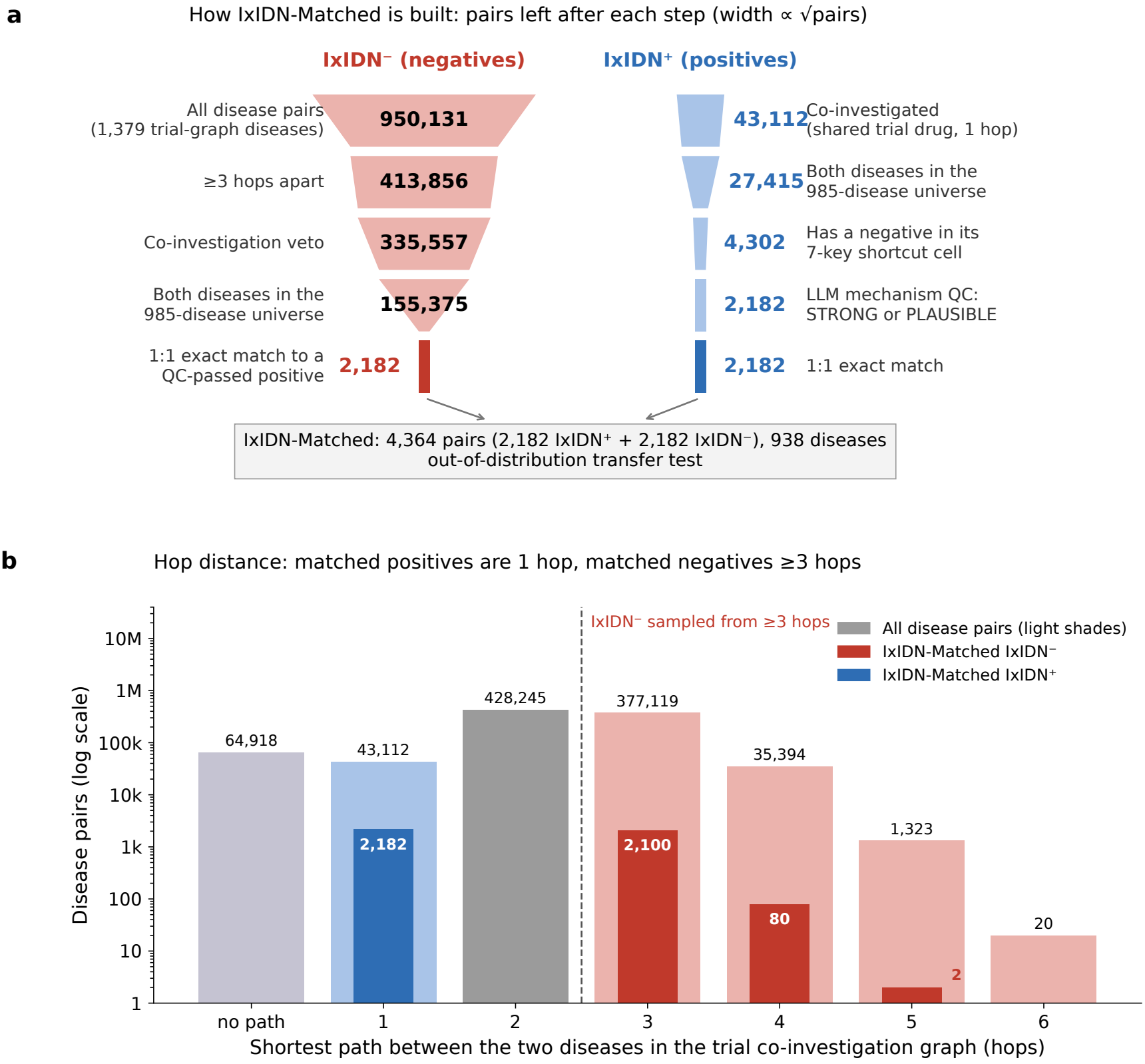
a
How IxIDN-Matched is built: pairs left after each step (width ∝ √pairs)
IxIDN⁻ (negatives)
IxIDN⁺ (positives)
All disease pairs (1,379 trial-graph diseases)
950,131
≥3 hops apart
413,856
Co-investigation veto
335,557
Both diseases in the 985-disease universe
155,375
1:1 exact match to a QC-passed positive
2,182
43,112
Co-investigated (shared trial drug, 1 hop)
27,415
Both diseases in the 985-disease universe
4,302
Has a negative in its 7-key shortcut cell
2,182
LLM mechanism QC: STRONG or PLAUSIBLE
2,182
1:1 exact match
IxIDN-Matched: 4,364 pairs (2,182 IxIDN⁺ + 2,182 IxIDN⁻), 938 diseases out-of-distribution transfer test
b
Hop distance: matched positives are 1 hop, matched negatives ≥3 hops
IxIDN⁻ sampled from ≥3 hops
All disease pairs (light shades)
IxIDN-Matched IxIDN⁻
IxIDN-Matched IxIDN⁺
Disease pairs (log scale)
10M
1M
100k
10k
1k
100
10
1
64,918
43,112
2,182
428,245
377,119
2,100
35,394
80
1,323
2
20
no path
1
2
3
4
5
6
Shortest path between the two diseases in the trial co-investigation graph (hops)